NOTES ON CONSTRUCTING $\mathbb{R}$

MATH 378

We begin by supposing we have all the standard precalculus properties of $\mathbb{Q}$.
Chapter 8, Definition * (Infinite Sequences). An *infinite sequence* with values in $\mathbb{Q}$ is a function whose domain is a set (called *the index set*) of the form $\{i \in \mathbb{Z} : i \geq n_0\}$, and whose codomain is $\mathbb{Q}$. We use notation such as $(a_i)_{i \geq n_0}$ to denote such a sequence. (Notice that we can always relabel so that the index set is $\mathbb{N}$.) We simply write (a_i) when the domain is not important to the discussion.
Chapter 8, Definition 11 (Limits). Suppose that (a_i) is a sequence. We say that L is *the limit of* (a_i) if the following holds: for all positive ε there is an $N \in \mathbb{N}$ such that if $i \geq N$ then $|a_i - L| < \varepsilon$.
Chapter 8, Definition 12 (Convergence). A sequence that has a limit is said to *converge*. We often employ the notation $\lim_{i\to\infty} a_i = L$. to indicate the sequence (a_i) converges to the limit L.
Chapter 8, Theorems 34 and 35. A field is Archimedean if and only if the following holds: if $u > 0$ is a positive element, then there is a positive $n \in \mathbb{N}$ such that $\frac{1}{n} \leq u$. Further, $\mathbb{Q}$ is Archimedean.
Theorem 38. A convergent sequence in an ordered field has a unique limit.

Proof. Suppose otherwise that (a_i) is a sequence in F with two distinct limits b and c. By trichotomy we get that $b < c$ or $c < b$. Assume without loss of generality that $b < c$. Let $\varepsilon = (c - b)/2$. We see $\varepsilon > 0$. By definition of limit, there is a $N_1 \in \mathbb{N}$ such that $|a_i - b| < \varepsilon$ for all $i \geq N_1$. Likewise there is a $N_2 \in \mathbb{N}$ such that $|a_i - c| < \varepsilon$ for all $i \geq N_2$. Let i be the maximum of N_1 and N_2. Then since $i \geq N_1$, $|a_i - b| < \varepsilon$. Hence $-\varepsilon < a_i - b < \varepsilon$. Using properties of ordered fields we get $a_i < b + \varepsilon = b + (c - b)/2 = (b + c)/2$. Next, since $i \geq N_2$, $|a_i - c| < \varepsilon$, a similar argument gives us $(b + c)/2 < a_i$. Putting these together we conclude $a_i < (b + c)/2 < a_i$, so $a_i < a_i$ by transitivity. This contradicts trichotomy. $\square$

Definition 13 (Equivalent sequences). Suppose that $(a_i)_{i\geq n_1}$ and $(b_i)_{i\geq n_2}$ are two sequences in an ordered field F. We write $(a_i) \sim (b_i)$ if the following occurs: for all positive $\varepsilon \in F$ there is a $N \in \mathbb{N}$ such that, for all $i \in \mathbb{N}$, $i \geq N \implies |a_i - b_i| < \varepsilon$. In other words, $(a_i) \sim (b_i)$ if $\lim_{i\to\infty} a_i - b_i = 0$.
Lemma 39. The relation $\sim$ is reflexive on the set of all sequences in F.
Lemma 40. The relation $\sim$ is transitive on the set of all sequences in F.
Lemma 41. The relation $\sim$ is symmetric on the set of all sequences in F. I.e., if $(a_i) \sim (b_i)$, then $(b_i) \sim (a_i)$.
Theorem 42. The relation $\sim$ is an equivalence relation on the set of sequences of (a_i) in F.
Theorem 43. Let (a_i) and (b_i) be sequences in an ordered field F and suppose $(a_i) \sim (b_i)$. If (a_i) has a limit, then (b_i) converges and has the same limit as (a_i).
Chapter 8, Theorem 44. Suppose $a_i = a$ for all i. Then $\lim_{i\to\infty} a_i = a$.
Chapter 8, Theorem 45. Suppose (a_i) and (b_i) are two converging sequences, and suppose there is a $N \in \mathbb{N}$ such that $a_i \leq b_i$ for all $i \geq N$. Then $\lim_{i\to\infty} a_i \leq \lim_{i\to\infty} b_i$.
Chapter 8, Theorem 47 (Sum law). If $\lim_{i\to\infty} a_i = a$ and $\lim_{i\to\infty} b_i = b$, then $\lim_{i\to\infty} a_i + b_i = a + b$.
Chapter 8, Theorem 48. If c is a constant and $\lim_{i\to\infty} a_i = a$, then $\lim_{i\to\infty} c \cdot a_i = c \cdot a$.
Chapter 8, Theorem 50 (Product law). If $\lim_{i\to\infty} a_i = a$ and $\lim_{i\to\infty} b_i = b$, then $\lim_{i\to\infty} a_i \cdot b_i = a \cdot b$.
Chapter 8, Theorem 53 (Inverse law). If $\lim_{i\to\infty} a_i = a \neq 0$, then $\lim_{i\to\infty} a_i^{-1} = a^{-1}$.

Definition 14 (Supremum). Let S be a nonempty subset of an ordered field F. A *supremum* M of S is a least upper bound in the following sense: $M \geq x$ for all $x \in S$, and if $B \in F$ is such that $B \geq x$ for all $x \in S$, then $M \leq B$.

Definition 15 (Infimum). Let S be a nonempty subset of an ordered field F. An *infimum* m of S is a greatest lower bound in the following sense: $m \leq x$ for all $x \in S$, and if $b \in F$ is such that $b \leq x$ for all $x \in S$, then $b \leq m$.

Chapter 8, Theorem 60 a.k.a. Chapter 9, Theorem 11. Let S be a nonempty subset of an Archimedean ordered field F. If M is the supremum of S, then M is the limit of a sequence (a_i) of elements $a_i \in S$ and is the limit of a sequence (b_i) of elements $b_i \notin S$. If $a < M$ is given we can assume that each a_i is in the interval $[a, M]$. Similarly, if $b > M$ is given we can assume that each b_i is in the interval $[M, b]$.

Proof idea: Choose a_i's in intervals like $(M - \frac{1}{i}, M]$. These will always contain an $a_i \in S$ since M is the supremum.

Chapter 9, Definition 1 (Completeness). Let F be an ordered field. We say that F is *complete* if every nonempty subset $S \subseteq F$ which is bounded from above has a supremum (least upper bound).

Chapter 9, Theorem 2. Suppose F is a complete ordered field containing $\mathbb{Q}$ as an ordered subfield. Then F is an Archimedean ordered field.

Proof. From results in Chapter 8 it is enough to show that for all $y \in F$ there is an $n \in \mathbb{N}$ such that $n \geq y$.

Suppose instead that this condition fails. Then there is a $y \in F$ such that $n < y$ for all $n \in \mathbb{N}$. This implies that $\mathbb{N}$ is bounded. Since F is complete, this means that $\mathbb{N}$ has a supremum M. Since $M - 1 < M$, the definition of supremum tells us that $M - 1$ is not an upper bound for $\mathbb{N}$, so there must be an integer $n \in \mathbb{N}$ such that $M - 1 < n$. From this we have that $M < n + 1$. Thus M is also not an upper bound of $\mathbb{N}$, a contradiction. $\square$

Chapter 9, Definition 2 (Continuity). Let S be a subset of an ordered field F. Then a function $f : S \to F$ is said to be *continuous* on S if for all converging sequences (a_i) with terms and limit in S, the sequence $\Big(f(a_i)\Big)$ also converges, and $\lim_{i\to\infty} f(a_i) = f\Big(\lim_{i\to\infty} a_i\Big)$.

Intuition: "I can draw the graph of f without picking my pencil up!"

Chapter 9, Definition 5 (Second Definition of Continuity). Let S be a subset of an ordered field F. Let $f : S \to F$ be a function and let $a \in S$. We say that f is *continuous at a* in the $\delta - \varepsilon$ sense if the following holds. For all $\varepsilon > 0$ in F there is a $\delta > 0$ such that for all $x \in S$ if $|x - a| < \delta$, then $|f(x) - f(a)| < \varepsilon$.

If f is continuous in this sense for all $a \in S$, then we say that $f : S \to F$ is *continuous on S* (in the $\delta - \varepsilon$ sense).

Chapter 9, Definition 6. (Sequential Continuity). Let S be a subset of an ordered field F. Let $f : S \to F$ be a function and let $a \in S$. We say that f *is continuous at a in the sequential sense* if the following holds. For all sequences (a_i) converging to a with terms in S, the sequence $f(a_i)$ converges to $f(a)$.

Chapter 9, Theorem 12 (Intermediate Value Theorem). Let F be a complete ordered field that contains $\mathbb{Q}$ as an ordered subfield. Let $[a, b]$ be a closed interval in F where $a < b$ are elements of F. Suppose $f : [a, b] \to F$ is continuous. If $C \in F$ is any value between $A = f(a)$ and $B = f(b)$, then there is an element $c \in [a, b]$ such that $f(c) = C$.

Proof. Without loss of generality we can assume $A < C < B$. (The case where $B < C < A$ is symmetric, and the case where C is A or B is trivial.) Define the following set

$$S \stackrel{\text{def}}{=} \{u \in [a, b] : f(u) \le C\}.$$

Observe that S is nonempty since $a \in S$ and that b is an upper bound for S. Since F is complete, the set S has a supremum, call it c. We claim that c satisfies the conclusion of the theorem.

We begin by showing $f(c) \le C$. By Theorem 11 there is a sequence (a_i) of elements in S that converges to c. We can construct such a sequence in the following manner. Let $a_0 = a$. For each $i \in \mathbb{N}$, choose $a_{i+1} \in [\frac{a_i+c}{2}, c] \cap S$. Such a choice exists since c is the least upper bound. We see $|c - a_i| = c - a_i \le \frac{c-a}{2^i}$. As i gets large, a_i gets arbitrarily close to c.) Since $a_i \in S$, we have $f(a_i) \le C$. Since f is continuous, the sequence $\big(f(a_i)\big)$ converges to $f(c)$. Thus $f(c) = \lim_{i\to\infty} f(a_i) \le C$.

Next we show that c is in the interval $[a, b)$. (For the sake of the theorem, we only need $c \in [a, b]$, but we need $c < b$ below in the proof.) Observe that $a \le c$ since $a \in S$ and c is an upper bound for S. Since b is an upper bound of S, we have $c \le b$ because c is the least upper bound of S. Finally, we have $b \ne c$ since $f(b) = B > C$ and $f(c) \le C$ (established above). Thus $a \le c < b$.

Finally we show $f(c) \ge C$, which with the above will yield the result. By Theorem 11 there is a sequence (b_i) of elements not in S converging to c, and since $c < b$ we can also assume that each $b_i \in [c, b]$. We can construct such a sequence in the following manner. Let $b_0 = b$. For each $i \in \mathbb{N}$, choose $b_{i+1} \in (c, \frac{b_i+c}{2}]$, where the open interval at c is merely to exclude potential elements of S. Such a choice of b_{i+1} exists since c is the least upper bound. We see $|b_i - c| = b_i - c \le \frac{b-c}{2^i}$. As i gets large, b_i gets arbitrarily close to c.)

In particular, each b_i is in the interval $[a, b]$ since $[c, b]$ is a subset of $[a, b]$. Since $b_i \notin S$ we have $f(b_i) \ge C$. (Actually $f(b_i) > C$, but we just need $\ge$). Since f is continuous, the sequence $\big(f(b_i)\big)$ converges to $f(c)$. Thus $f(c) = \lim_{i\to\infty} f(b_i) \ge C$. □

Chapter 9, Corollary 13. Suppose $C \in F$ where F is a complete ordered field that contains $\mathbb{Q}$ as an ordered subfield. If $C \ge 0$, then there is an element $c \in F$ such that $c^2 = C$.

Chapter 9, Corollary 14. The field $\mathbb{Q}$ is not complete. Proof sketch: $\sqrt{2} \notin \mathbb{Q}$

Chapter 9, Definition 7 (Cauchy Sequences). Suppose (a_i) is an infinite sequence in an ordered field F. We say that (a_i) is *Cauchy* if the following occurs: for all $\varepsilon > 0$ in F there is a $N \in \mathbb{N}$ such that for all $i, j \in \mathbb{N}$,

$$i, j \ge N \implies |a_i - a_j| < \varepsilon.$$

Chapter 9, Theorem 15. Suppose (a_i) is a convergent sequence in an ordered field F. Then for all $\varepsilon > 0$ in F there is an $N \in \mathbb{N}$ such that for all $i, j \in \mathbb{N}$

$$i, j \ge N \implies |a_i - a_j| < \varepsilon.$$

Chapter 9, Theorem 16. If $(a_i) \sim (b_i)$ and if (a_i) is Cauchy, then (b_i) is Cauchy.

Chapter 9, Theorem 18 (The Cauchy Criterion for Completeness). Let F be an Archimedean ordered field. If every Cauchy sequence in F converges in F, then F is complete.

Theorem (Equivalent Conditions for the Completeness of $\mathbb{R}$). For $\mathbb{R}$ the following are equivalent:

(1) Every Cauchy sequence converges. (My preferred definition of completeness.)
(2) Every nonempty bounded subset has a supremum. (The book's definition of completeness.)
(3) Every continuous function satisfies the intermediate value property. (In other words, the intermediate value theorem holds for every continuous function.)

Proof. We have already seen (2) implies (3) from Theorem 12, the intermediate value theorem.

(2) $\implies$ (1): Let (a_i) be Cauchy. Then it is bounded, so we can consider the tails $s_n = \sup\{a_i : i \geq n\}$. Each s_n exists by our hypothesis that (2) holds. Let $L = \inf_n s_n$. This again exists by hypothesis. Note $\inf S = -\inf -S$. We claim $\lim_{i\to\infty} a_i = L$. To see this, fix $\varepsilon > 0$. Define $\mathbf{i}_n = \inf\{a_i : i \geq n\}$. Since (a_i) is Cauchy, we can choose $N \in \mathbb{N}$ so that if $i, j \geq N$, then $|a_i - a_j| < \varepsilon'$. Thus if $n \geq N$, then every term in $\{a_i : i \geq n\}$ is in an interval of length $2\varepsilon'$. By definition of sup and inf, we have $\mathbf{i}_n \leq a_n \leq s_n$. Again, since (a_i) is Cauchy, $s_n - \mathbf{i}_n \leq \varepsilon'$. By construction, $\mathbf{i}_n \leq L \leq s_n$. Thus, $|a_n - L| \leq s_n - \mathbf{i}_n \leq \varepsilon'$. Choosing an $\varepsilon' < \varepsilon$, we have our result.

(3) $\implies$ (2): Let S be a nonempty set that is bounded above by b. For a contradiction, suppose S has no supremum. Let $a \in S$, and define a function $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = 1$ if x is not an upper bound for S and $f(x) = -1$ if x is an upper bound for S. To see that f is continuous let (a_i) be a sequence in $\mathbb{R}$ such that $\lim_{i\to\infty} a_i = L \in \mathbb{R}$. If L is not an upper bound for S, then there exists $s \in S$ such that $s > L$. Let $\varepsilon = s - L > 0$. From the definition of convergence, there exists $N \in \mathbb{N}$ such that if $i \geq N$, then $|a_i - L| < \varepsilon$. Thus if $i \geq N$, then a_i is not an upper bound for S, so $f(a_i) = 1$. Hence $f\Big(\lim_{i\to\infty} a_i\Big) = \lim_{i\to\infty} f(a_i) = 1$. Conversely, suppose L is an upper bound for S. Since S does not have a supremum, there exists $r < L$ such that r is an upper bound for S. Let $\varepsilon = L - r$. From the definition of convergence, there exists $N \in \mathbb{N}$ such that if $i \geq N$, then $|a_i - L| < \varepsilon$. Thus if $i \geq N$, then a_i is an upper bound for S, so $f(a_i) = -1$. Hence $f\Big(\lim_{i\to\infty} a_i\Big) = \lim_{i\to\infty} f(a_i) = -1$. We have verified that f is continuous. The intermediate value property shows that there exists a $c \in [a, b]$ such that $f(c) = 0 \in [-1, 1]$. By construction this is not possible, so we have our desired contradiction.

(1) $\implies$ (2): Let S be a nonempty set that is bounded above. Let $a_0 \in S$ and b_0 and upper bound for S. We will construct a sequence a_i of larger and larger elements of S and a sequence b_i of smaller and smaller upper bounds for S. Consider $m_0 = \frac{a_0+b_0}{2}$. If m_0 is an upper bound for S, set $a_1 = a_0$ and $b_1 = m_0$. If m_0 is not an upper bound for S, choose a_1 to be an element of S such that $a_1 \geq m_0$ and take $b_1 = b_0$. Continue in this manner with $m_i = \frac{a_i+b_i}{2}$, $a_{i+1} = a_i$ and $b_{i+1} = m_i$ if m_i is an upper bound for S, and a_{i+1} an element of S such that $a_{i+1} \geq m_i$ and $b_{i+1} = b_i$ if m_i is not an upper bound for S. We see that for every i, $a_i \leq a_{i+1} \leq b_{i+1} \leq b_i$ and $b_i - a_i = \frac{b_0 - a_0}{2^i}$. Thus (a_i) and (b_i) are both Cauchy and they are equivalent, $\lim_{i\to\infty} b_i - a_i = 0$. Hence, we have $\lim_{i\to\infty} a_i = \lim_{i\to\infty} b_i = L$ for some $L \in \mathbb{R}$. We claim L is the supremum of S. Indeed, if L is not an upper bound, then there exists $s \in S$ with $s > L$. However, each $b_i \geq s$, so we have a contradiction. If L is not the least upper bound, then there exists an upper bound $B \in \mathbb{R}$ with $B < L$. But, each $a_i \leq B$, and we have a contradiction. Thus every nonempty set that is bounded above has a supremum. $\square$

Chapter 10, Definition 1 (A Real Number). If (a_i) is a Cauchy sequence in $\mathbb{Q}$, then let $\left[(a_i)\right]$ be the equivalence class containing (a_i) under the equivalence relation $\sim$ on the set of sequences in $\mathbb{Q}$. We call $\left[(a_i)\right]$ a *real number.*

Chapter 10, Definition 2 (The Real Numbers). The set of real numbers $\mathbb{R}$ is defined as follows:

$$\mathbb{R} \stackrel{\text{def}}{=} \left\{\left[(a_i)\right] : (a_i) \text{ is a Cauchy sequence in } \mathbb{Q}\right\}$$

Chapter 10, Definition 3 (Addition and Multiplication). Let $\left[(a_i)\right]$ and $\left[(b_i)\right]$ be real numbers. Then

$$\left[(a_i)_{i\geq n_0}\right] + \left[(b_i)_{i\geq m_0}\right] = \left[(a_i + b_i)_{i\geq \max(n_0,m_0)}\right]$$

and

$$\left[(a_i)_{i\geq n_0}\right] \cdot \left[(b_i)_{i\geq m_0}\right] = \left[(a_i \cdot b_i)_{i\geq \max(n_0,m_0)}\right].$$